

Revised Generation Interconnection System Impact Study Report

For Queue Project Q20-1479

Monfort 230 kV
June 29, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1479. The Interconnection Customer proposes a 358.9 MW wind facility interconnecting at the Monfort 230 kV substation in the Indiana area. The request was received on May 16, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$21,006,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Portage Uppco to Columbia Hills 100 kV	Waynedale to Monfort 230 kV	198.6	143	11.1
Illinois Rd to Southport 100 kV	Columbia Hills to Indego 100 kV	193.6	127	5.4
Waynedale to Indego 100 kV	Waynedale to Monfort 230 kV L	181.2	242	13.5
Hart to Southport 100 kV	Columbia Hills to Indego 100 kV	176.5	152	5.4
Portage Uppco to Hart Substation 100 kV	Columbia Hills to Indego 100 kV	164.3	177	5.4
Columbia Hills to Indego 100 kV	Monfort to Waynedale 230 kV L	159.0	294	13.4
Peavine to Hohenwald 100 kV line	Waynedale to State St 230 kV L	154.5	147	7.2
Illinois Rd to Hebron Rosemeade 100 kV	Waynedale to Indego 100 kV	152.1	163	6.1
Mckinley to Hebron Rosemeade 100 kV	Waynedale to Monfort 230 kV	139.8	134	5.5
Tiefert to Winder 100 kV Line 1	Waynedale Substation to Monfort 230 kV	131.3	151	9.1
Tiefert to Winder 100 kV	Waynedale to Monfort 230 kV	131.3	151	9.1
Portage Uppco to Tiefort 100 kV line	Mnfort to Waynedale 230 kV	129.4	151	9.1
Portage Uppco to Tiefort 100 kV Circuit 1	Waynedale to Monfort 230 kV Line	126.1	154	9.1
Wayne Ic to Wrangell 100 kV	Monfort to Wabash 230 kV	124.8	145	6.9
Waynedale to Monfort 230 kV	Wabash 500/230 kV Auto	120.9	1,323	72.1
Spy Run to Industrial Park 100 kV	Waynedale to State St Substation 230 kV	120.1	161	5.4
Monfort to Winder 100 kV #2	Waynedale to Monfort 230 kV	116.4	182	9.1

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Segs 5 Substation to Aep Pettit Avenue 100 kV line	State Street to Waynedale 230 kV	109.8	265	7.2
Monfort to Winder 100 kV #1	Monfort to Waynedale Substation 230 kV	108.1	195	9.1
Wayne Ic to Deercrk 100 kV Line	Wabash to Monfort 230 kV	107.3	154	6.9
Waynedale to Aep Pettit Avenue 100 kV Cir. 1	Waynedale to Aep Pettit Avenue 100 kV Circuit 2	101.2	340	6.7
Waynedale to Aep Pettit Avenue 100 kV Ckt 2	Waynedale to Aep Pettit Avenue 100 kV	101.2	340	6.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Columbia Hills to Portage Uppco 100 kV	Rebuild line with higher-capacity conductor	\$3,887,000	13% (\$503,000)
Illinois Rd to Southport 100 kV	Rebuild line with higher-capacity conductor	\$4,649,000	8% (\$357,000)
Waynedale to Indego 100 kV	Rebuild line with higher-capacity conductor	\$4,052,000	12% (\$484,000)
Hart to Southport 100 kV	Rebuild line with higher-capacity conductor	\$2,061,000	8% (\$158,000)
Portage Uppco to Hart 100 kV	Rebuild line with higher-capacity conductor	\$4,515,000	8% (\$347,000)
Indego to Columbia Hills 100 kV	Rebuild line with higher-capacity conductor	\$2,168,000	12% (\$256,000)
Peavine to Hohenwald 100 kV line	Rebuild line with higher-capacity conductor	\$2,255,000	12% (\$276,000)
Illinois Rd to Hebron Rosemeade 100 kV	Rebuild line with higher-capacity conductor	\$1,854,000	7% (\$134,000)
Mckinley to Hebron Rosemeade 100 kV	Rebuild line with higher-capacity conductor	\$3,171,000	11% (\$342,000)
Tiefert Substation to Winder 100 kV #1	Rebuild line with higher-capacity conductor	\$10,125,000	20% (\$2,016,000)
Tiefert to Winder 100 kV	Rebuild line with higher-capacity conductor	\$3,831,000	20% (\$763,000)
Portage Uppco to Tiefort 100 kV Line 2	Rebuild line with higher-capacity conductor	\$13,223,000	20% (\$2,633,000)
Portage Uppco to Tiefort 100 kV Circuit 1	Rebuild line with higher-capacity conductor	\$7,947,000	20% (\$1,582,000)
Wayne Ic Substation to Wrangell 100 kV	Rebuild line with higher-capacity conductor	\$2,334,000	15% (\$346,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Waynedale to Monfort 230 kV	Rebuild line with higher-capacity conductor	\$43,801,000	21% (\$9,367,000)
Spy Run to Industrial Park 100 kV	Rebuild line with higher-capacity conductor	\$1,177,000	10% (\$124,000)
Monfort to Winder 100 kV #2	Rebuild line with higher-capacity conductor	\$3,767,000	20% (\$750,000)
Segs 5 to Aep Pettit Avenue 100 kV Line	Reconductor line	\$671,000	13% (\$86,000)
Monfort to Winder 100 kV Ckt 1	Reconductor line	\$1,151,000	20% (\$229,000)
Wayne Ic to Deer Crk 100 kV	Reconductor line	\$638,000	15% (\$94,000)
Waynedale to Aep Pettit Avenue 100 kV Ckt 1	Reconductor line	\$512,000	12% (\$61,000)
Waynedale to Aep Pettit Avenue 100 kV Ckt 2	Reconductor line	\$824,000	12% (\$98,000)
	Total allocated to project		\$21,006,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.